Drone flying and operations

ConOps and risk, flight modes, control, safe operations, monitoring, DGCA (Appendix 5B)

ConOps and risk

- ConOps: who, what, where, when, how, limits, what if.
- Risk R = S × L. Controls reduce S or L; record the residual risk.
- SORA: ground risk class + air risk class → SAIL → operational safety objectives.

Flight modes (more automation →)

- Rate/Acro → Stabilize → AltHold (+ baro) → Position hold (+ GNSS, compass) → Mission/Guided → Return → Land.
- Each step adds a sensor and fails when that sensor fails.
- RTL needs: valid home, altitude above obstacles, energy against the wind, healthy heading and position.
- Failsafe priority by severity: critical battery or attitude-estimate loss → LAND, ahead of fence or link loss.

Dynamics and control

- Tilt to translate: $a = g \cdot \tan\varphi$; hold altitude: $T = mg/\cos\varphi$.
- Cascade: position → velocity → attitude → rate → mixer; inner loops 3–10× faster; tune inside out.
- Kp ↑ faster, more overshoot. Ki ↑ removes steady error, adds windup. Kd ↑ damping, amplifies noise.
- Windup cure: clamping or conditional integration. Ramp error $e_{ss} = 1/K_v$.
- Delay τ costs phase margin $\omega_c \cdot \tau$. Quadcopters are open-loop unstable.

Safe operations

- Crew: remote pilot (in command), visual observer, GCS/payload operator.
- Go/no-go limits are numbers: wind, satellites and HDOP, battery, area clear.
- First flight: config record → props-off bench → tethered → low hover → modes → failsafes low → expand envelope.
- Return time uses ground speed: $t = d/(V_a \pm V_w)$; plan into the headwind, then add reserve.

Monitoring and logs

- Logs: .bin (ArduPilot), .ulg (PX4), .tlog (GCS telemetry).
- Diagnose from time-aligned logs: vibration ↑ → EKF innovations ↑ → altitude error → motor saturation → altitude loss.
- Trend hover current, vibration, motor temperature, pack resistance, cycle count.

DGCA (Drone Rules 2021, as amended)

- Nano, micro, small, medium, large. UIN on Digital Sky; type certificate (with exemptions).
- RPC from a DGCA-authorized RPTO; not needed for nano, or micro flown non-commercially.
- Green: no permission, up to 400 ft AGL. Yellow: ATC permission. Red: Central Government only.
- Check the Digital Sky map before every flight.

Exam traps

- "RTL is always safe." Raising Kp to fix a saturated axis.
- Return time from airspeed. Quoting US Part 107 as Indian rules.
- Skipping checklists and reserves in a "controlled" area.

Source: course reader, Appendix 5B